

In C++, `goto` or `switch` cannot cross an initialisation. The program *Example 12* works fine as a C program but fails to compile as a C++ program. In this program, `goto` is crossing an initialisation, which is an error in C++. The error message *crosses initialization of 'int i'* is displayed when the program is treated as C++.

```
#include<stdio.h>
int main()
{
    goto jump_over_initialization;
    int i=0;
    printf("\nI will not be printed\n");
    jump_over_initialization:
}
```

Example 12

Similarly, refer the program *switch.c*, in which the error is due to the fact that `switch` is crossing an initialisation. In C, an enum variable can take any value, whereas in C++, only the values specified in the declaration can be assigned to an enum variable. In C, a character array overflow is tolerated with a warning, whereas C++ is stricter about this matter and will directly lead to a compile time error. But, of course, the option `-Werror` of gcc, if enabled, will result in a compile time error for the program *array.c* in C also. The result of a Boolean expression is different in C and C++. The program *Example 13* will print 4 as the output when treated as a C program and 1 as the output when treated as a C++ program.

```
#include<stdio.h>
int main()
{
    printf("\n%d\n", sizeof(i==1));
}
```

}

Example 13

The size of a structure with no members is different in C and C++. The program *Example 14* gives 0 as the output when treated as a C program and 1 as the output when treated as a C++ program.

```
#include<stdio.h>
struct empty
{
};
int main()
{
    printf("\n%d\n", sizeof(struct empty));
}
```

Example 14

I hope these minor differences between C and C++ pointed out in this article will help students and computer professionals in technical interviews. But do remember that this article does not cover all such differences and I will be much obliged if somebody could point out any other such differences that are not covered in this article. Before concluding, I would like to acknowledge my brilliant student, Julin Shaji, for identifying many of these subtle differences between C and C++. 

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